

## Project Design Phase-II

### Solution Requirements (Functional & Non-Functional)

|               |                                                        |
|---------------|--------------------------------------------------------|
| Date          | 21 February 2026                                       |
| Team ID       | LTVIP2026TMIDS42560                                    |
| Project Name  | Online Payments Fraud Detection using Machine Learning |
| Maximum Marks | 4 Marks                                                |

| NFR No. | Non-Functional Requirement | Description                                                                                                    |
|---------|----------------------------|----------------------------------------------------------------------------------------------------------------|
| NFR-1   | Usability                  | Provide a simple and intuitive interface for customers and fraud analysts with clear fraud/legitimate results. |

|       |          |                                                                           |
|-------|----------|---------------------------------------------------------------------------|
| NFR-2 | Security | Ensure secure transaction data handling using encryption, authentication, |
|-------|----------|---------------------------------------------------------------------------|

### Functional Requirements

| FR No. | Functional Requirement (Epic)      | Sub Requirement (Story / Sub-Task)                                                    |
|--------|------------------------------------|---------------------------------------------------------------------------------------|
| FR-1   | Transaction Input & Validation     | Accept transaction details (amount, time, location, device info) and validate inputs  |
| FR-2   | Fraud Prediction                   | Classify transaction as Fraud or Legitimate using ML model                            |
| FR-3   | Data Preprocessing                 | Handle missing values, encode categorical features, scale numerical inputs            |
| FR-4   | Model Inference                    | Load trained fraud detection model and return prediction with fraud probability score |
| FR-5   | Alert & Notification System        | Trigger alert/notification for suspicious transactions                                |
| FR-6   | Dashboard & Reporting              | Display fraud statistics, trends, and transaction monitoring dashboard                |
| FR-7   | Dataset & Model Management (Admin) | Upload/update transaction datasets and retrain or replace ML model                    |

### Non-Functional Requirements

|       |             |                                                                                                                |
|-------|-------------|----------------------------------------------------------------------------------------------------------------|
|       |             | and strict input validation.                                                                                   |
| NFR-3 | Reliability | Maintain consistent preprocessing and prediction using saved model artifacts (encoder, scaler, trained model). |
| NFR-4 | Performance | Deliver real-time fraud detection with low latency (less than 1 second per transaction).                       |

|       |                 |                                                                                      |
|-------|-----------------|--------------------------------------------------------------------------------------|
| NFR-5 | Availability    | System should operate 24/7 with minimal downtime to support continuous transactions. |
| NFR-6 | Scalability     | Handle high volumes of concurrent transactions without performance degradation.      |
| NFR-7 | Maintainability | Use modular and well documented code to allow easy updates and model retraining.     |
| NFR-8 | Portability     | Support deployment on Windows/Linux systems and be cloud-ready for future scaling.   |